

BASICS OF FIM · WI001287

Statistical Learning

4: Logistic regression & Bayes' classification

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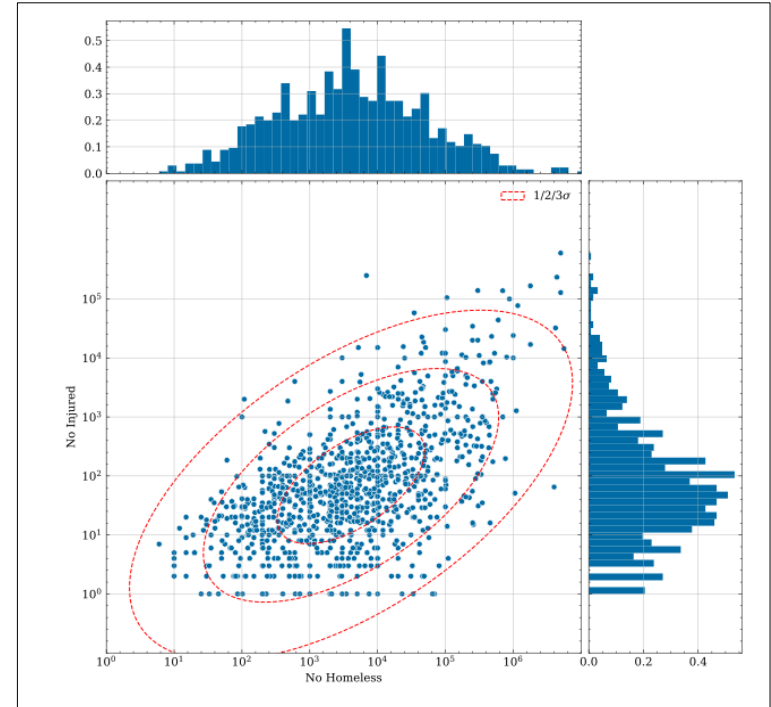


Figure 3.3: Residual sum squared loss surface

- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

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Recap: Lecture 2 Regression vs. Classification

Measurable Machine Learning Problems



Business Problem

Objectives and business success criteria describe the problem. However, these descriptions are often subjective and lack measurability.



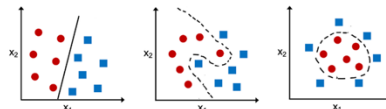
Translation

Machine Learning Problem

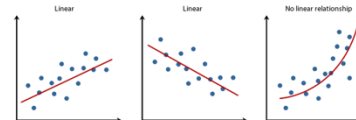
Machine learning algorithms on the other hand require a **clear, measurable goal** that can be quantified by a dominate metric.



Classification | classify instances into classes



Regression | predict numerics as close as possible



Recap: Lecture 3 introduced linear regression

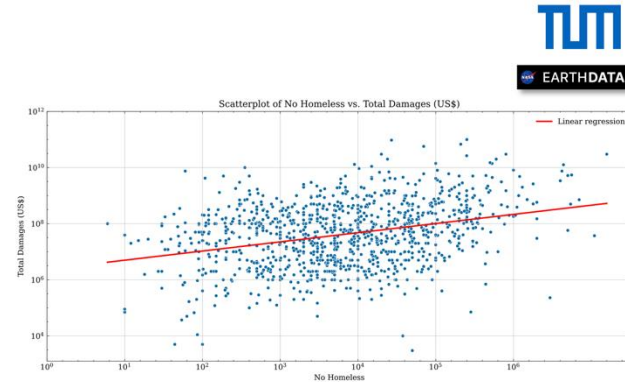
We bet the world is a line plus noise.

Linear regression is a simple approach to supervised learning. We believe about the world that: $Y \approx f(X) + \varepsilon$. In the case of linear regression, we assume that the dependence of Y on X is linear:

$$Y = \beta_0 + \beta_1 \times X + \varepsilon$$

$\swarrow$ y-axis intercept
 $\swarrow$ error term
 $\nwarrow$ slope

... but we can not observe this entity
 ... instead we have a finite set of observations $\{(x_i, y_i)\}_{i=1}^N$
 ... from which we learn the function:
 $\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$



True regression functions are never linear!

Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

What if we want to predict the type of disaster?

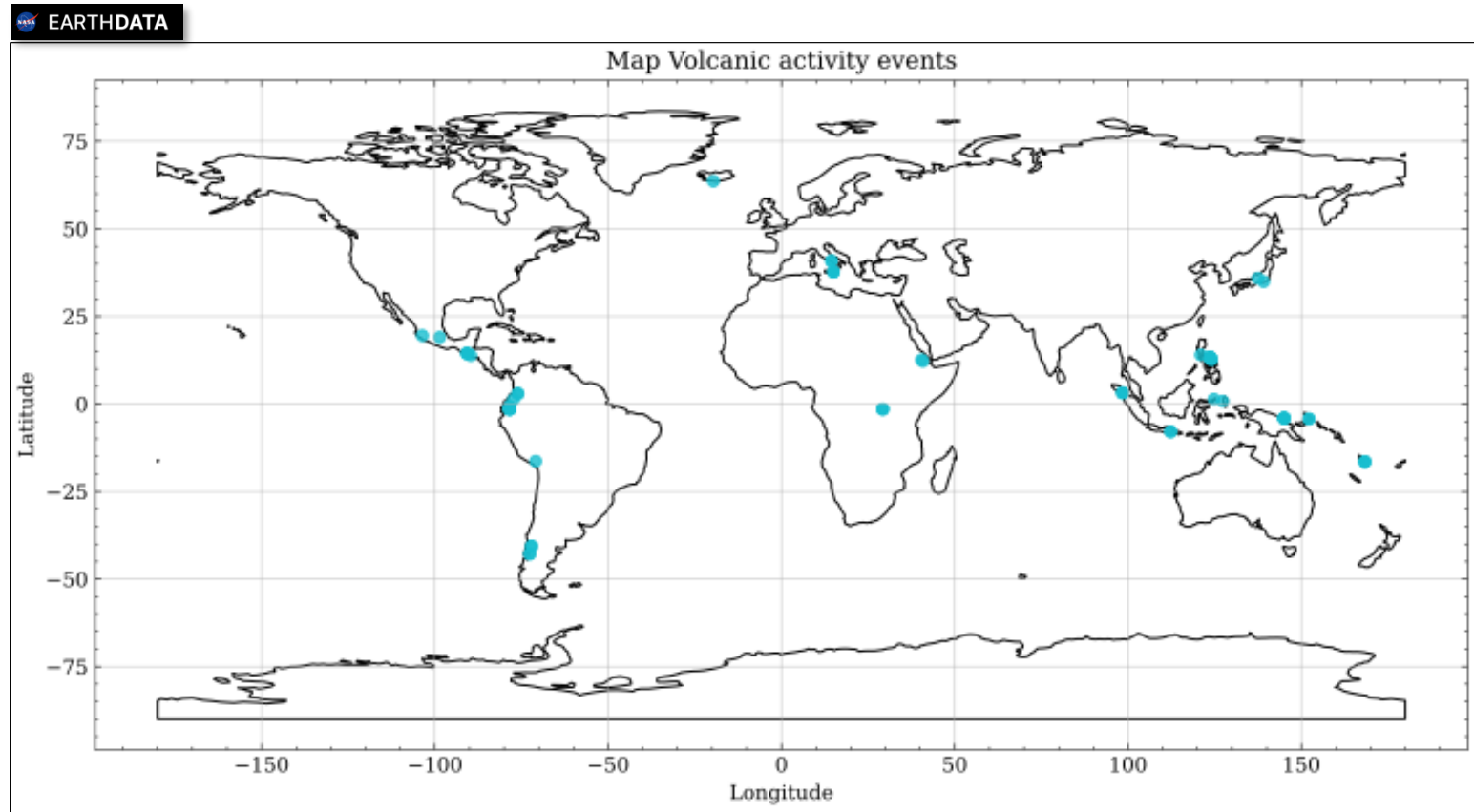
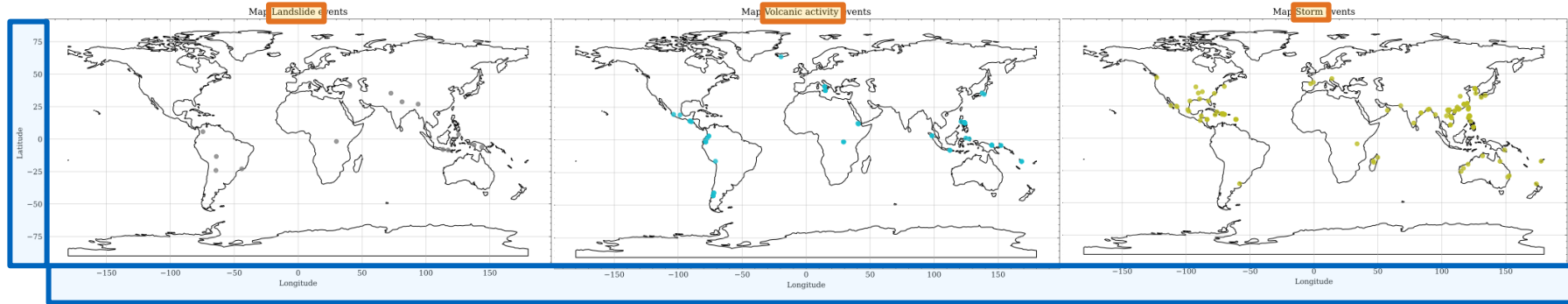


Figure 4.1: world map

What if we want to predict the type of disaster?



Feature: an individual measurable property or characteristic of a phenomenon

Target: is the variable the model is trying to predict

Can we use linear regression to predict disaster types?

EARTHDATA

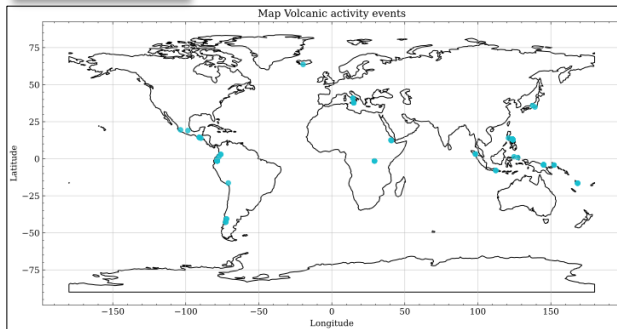


Figure 4.1: world map

Let's define our classification task as:

$$Y = \begin{cases} 1 & \text{if volcanic activity} \\ 0 & \text{else} \end{cases}$$

Can we simply perform a **linear regression** of Y on X and classify as volcanic activity if $\hat{Y} > 0.5$?



Does the linear model fit our task?

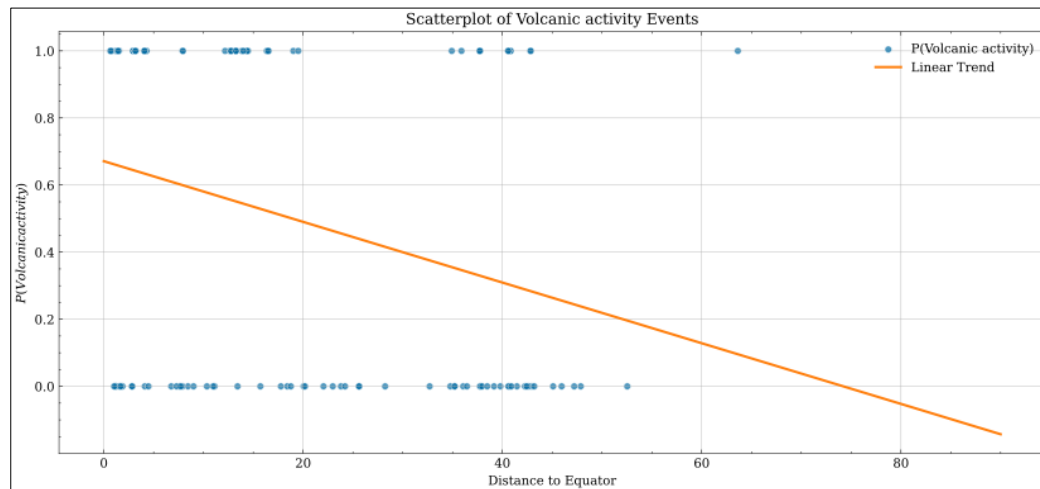


Figure 4.2: linear attempt

Can we use linear regression to predict disaster types?

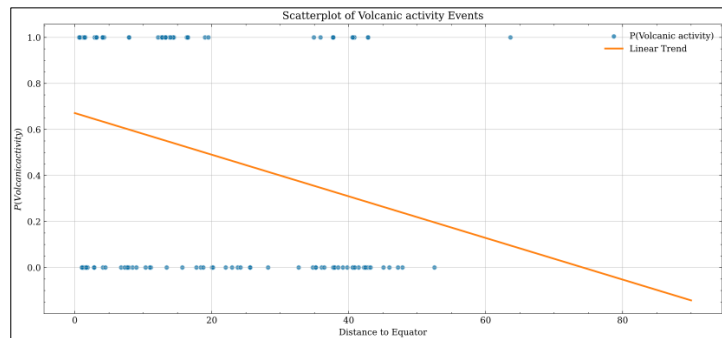


Figure 4.2: linear attempt

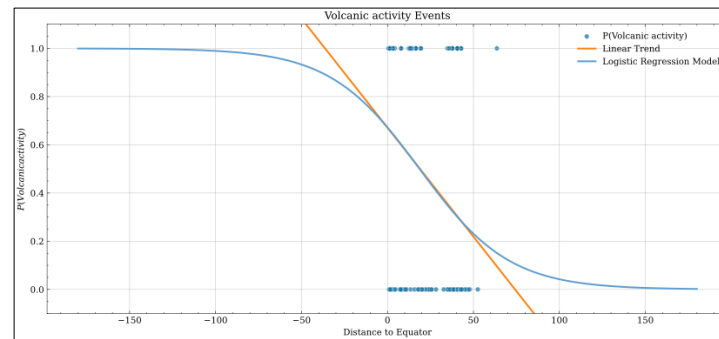


Figure 4.3: logistic fit

We want to ensure that: $P(X) \in [0,1]$

→ We need a link function that relates the linear model to the parameter of the outcome distribution.

Logistic transformation ensures $P(X) \in [0,1]$.

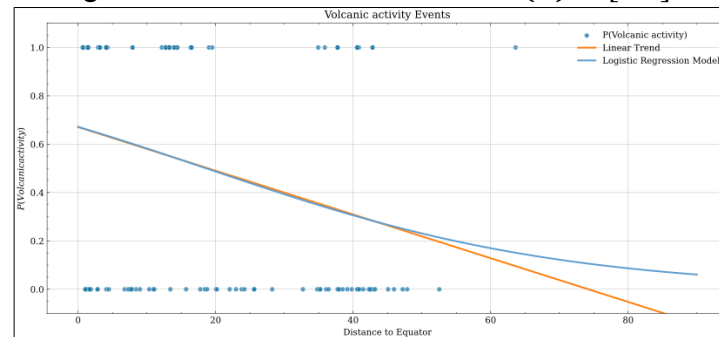


Figure 4.3: logistic fit



Is logistic regression a better fit?

Agenda.



- Recap on classification and regression
- **Logistic regression**
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

Logistic regression ensures $P(X) \in [0,1]$

EARTHDATA

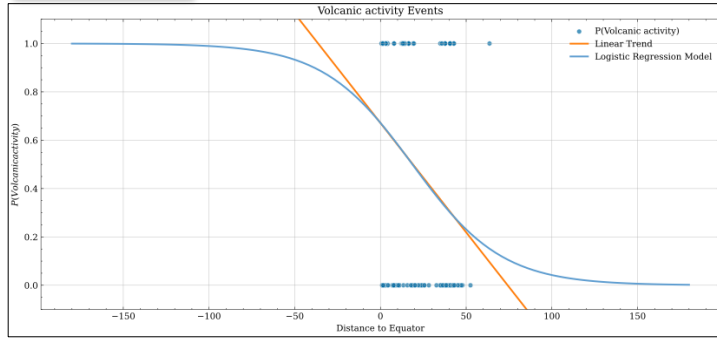


Figure 4.3: logistic fit

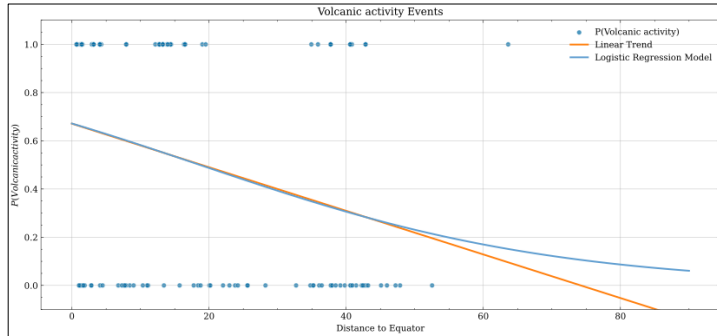


Figure 4.3: logistic fit

Let's write $P(X) = P(Y = 1|X)$ for short. Further apply logit transformation as follows:

$$P(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

It's easy to see that no matter what values β_0 , β_1 or X take $P(X)$ will have values between 0 and 1.

A bit of rearrangement gives:

$$\log\left(\frac{P(X)}{1 - P(X)}\right) = \beta_0 + \beta_1 X$$

This monotone transformation is called the log odds or logit transformation of $P(X)$.

We use maximum likelihood to estimate the parameters.

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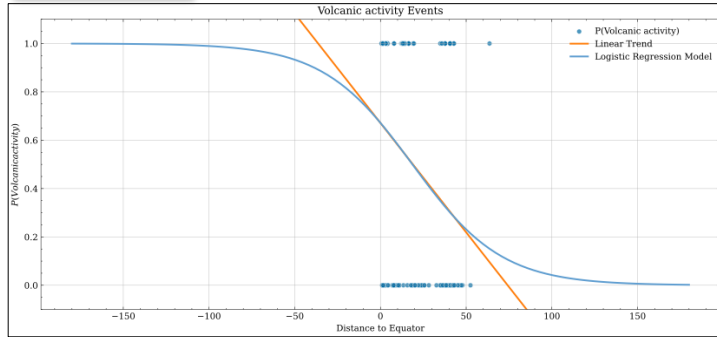


Figure 4.3: logistic fit

Logistic regression:

$$P(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

We pick β_0 and β_1 to maximize the likelihood of the observed data:

$$\mathcal{L}(\beta_0, \beta_1) = \prod_{i:y_i=1} P(x_i) \times \prod_{i:y_i=0} (1 - P(x_i))$$

Logit Regression Results

Dep. Variable:	P(Volcanic activity)	No. Observations:	124
Model:	Logit	Df Residuals:	122
Method:	MLE	Df Model:	1
Date:	Fri, 17 Oct 2025	Pseudo R-squ.:	0.06352
Time:	23:17:18	Log-Likelihood:	-80.491
converged:	True	LL-Null:	-85.950
Covariance Type:	nonrobust	LLR p-value:	0.0009522

	coef	std err	z	P> z	[0.025	0.975]
const	0.7163	0.290	2.474	0.013	0.149	1.284
Distance to Equator	-0.0384	0.012	-3.164	0.002	-0.062	-0.015



How do we interpret the results?

Logistic regression also works for multiple predictors and multiple target classes.

Multiple Logistic Regression

We now consider the problem of predicting a binary response using multiple predictors.

ID	Feature 1	...	Feature p	Target
1				Yes
2				No
3				No
4				Yes
5				No
...				Yes
n				Yes

By analogy to multiple linear regression, we can generalize as follows:

$$\log\left(\frac{P(X)}{1 - P(X)}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

And we can rewrite to:

$$P(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}$$

Multinomial Logistic Regression

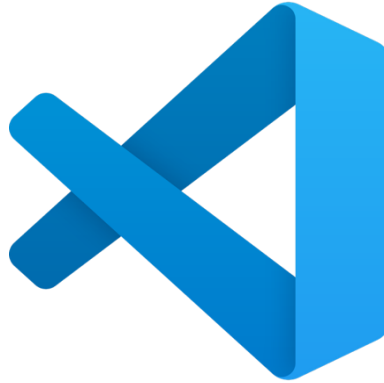
And we can consider response variables with more than two classes:

ID	Feature 1	...	Feature p	Target
1				Flood
2				Landslide
3				Storm
4				Drought
5				Storm
...				Epidemic
n				Wildfire

$$P(Y = k|X) = \frac{e^{\beta_{0k} + \beta_{1k}X_1 + \dots + \beta_{pk}X_p}}{\sum_{l=1}^K e^{\beta_{0l} + \beta_{1l}X_1 + \dots + \beta_{pl}X_p}}$$

Feature: an individual measurable property or characteristic of a phenomenon

Target: is the variable the model is trying to predict



Agenda.



- Recap on classification and regression
- Logistic regression
- **Linear discriminant analysis**
- Quadratic discriminant analysis
- Naïve Bayes

Discriminant analysis take the opposite perspective on conditional distributions.

Logistic Regression directly models $P(Y = k|X)$, statistically we say we model the conditional distribution of the response Y , given the predictor(s) X .

By contrast **discriminant analysis** start with the conditional distribution of X given Y and then use **Bayes formula** to turn things around.

$$P(Y = k|X = x) = \frac{P(X = x|Y) \times P(Y = k)}{P(X = x)}$$

Why do we need discriminant when we have logistic regression?

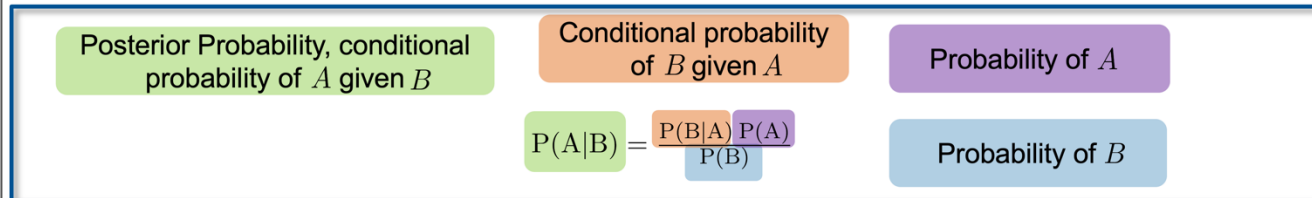
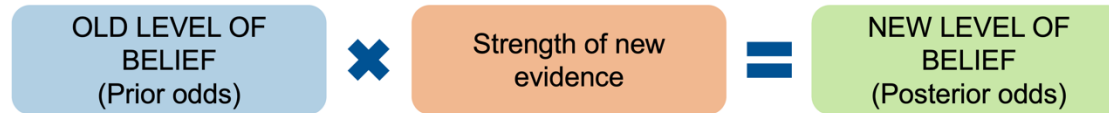
- When the classes are well-separated, the parameter estimates for the logistic regression model are surprisingly unstable. Linear discriminant analysis does not suffer from this problem.
- If n is small and the distribution of the predictors X is approximately normal in each of the classes, the linear discriminant model is again more stable than the logistic regression model.
- Linear discriminant analysis is popular when we have more than two response classes, because it also provides low-dimensional views of the data.

Recap: Lecture 2 Bayes theorem

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Bayes Theorem - Motivation



Bayes Theorem provides a way to update the probability of a hypothesis based on new evidence. The theorem considers both the initial probability of the hypothesis and how likely the new evidence is.

Define priors and density functions for discriminative analysis.

Discriminant analysis start with the conditional distribution of X given Y and then use **Bayes formula** to turn things around.

$$P(Y = k|X = x) = \frac{P(X = x|Y) \times P(Y = k)}{P(X = x)}$$

In **discriminant analysis** we write

$$P(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}, \text{ where}$$

$f_k(x) = P(X = x|Y)$ is the density for X in class k . Here we will use normal densities for these, separately in each class.

$\pi_k = P(Y = k)$ is the marginal or prior probability for class k .

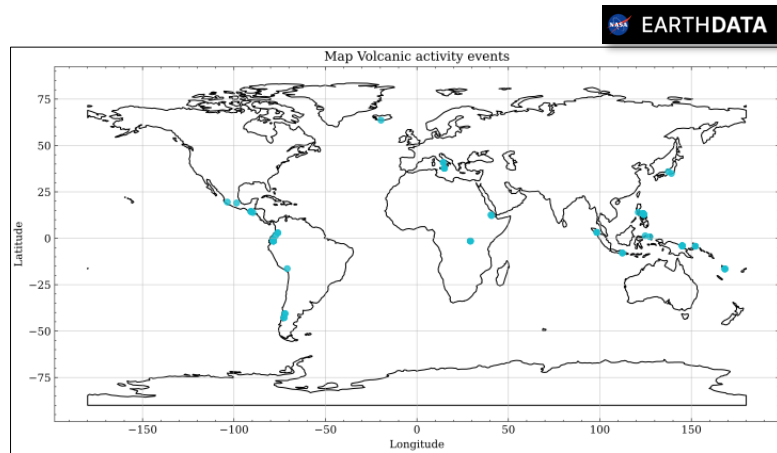


Figure 4.1: world map

ID	Distance Equator	Volcanic
1	...	Yes
2	...	No
...	...	Yes
n	13°	Yes

Total
 $h_s(f_s)$ 63
(0.03)

We assume priors in discriminative analysis to be gaussian.

Discriminant analysis

$$P(Y = k|X = x) = \frac{\pi_k \times f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

The **gaussian density** function has the form:

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}, \text{ with}$$

μ_k is the mean and σ_k^2 the variance (in class k).

We will assume that all $\sigma_k = \sigma$ are the same.



How strong is this assumption?

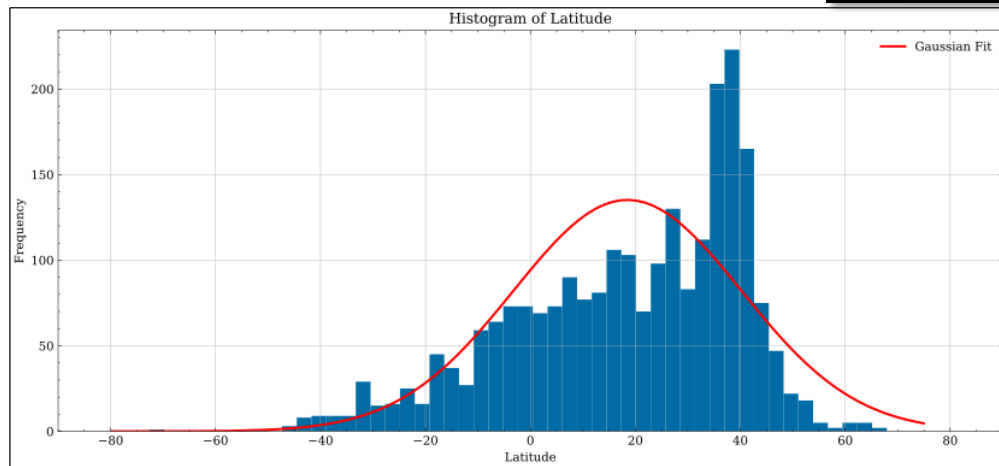


Figure 4.4: class prior

We get:

$$P(Y = k|X = x) = \frac{\pi_k \times \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma}\right)^2}}{\sum_{l=1}^K \pi_l \times \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_l}{\sigma}\right)^2}}$$

We can break down how to estimate parameters and run inference.

Discriminant analysis

$$P(Y = k|X = x) = \frac{\pi_k \times \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma}\right)^2}}{\sum_{l=1}^K \pi_l \times \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_l}{\sigma}\right)^2}}$$

We **estimate parameters** as follows:

$$\hat{\pi}_k = \frac{n_k}{n}$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i=k} x_i$$

$$\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^K \sum_{i: y_i=k} (x_i - \hat{\mu}_k)^2$$

Running inference

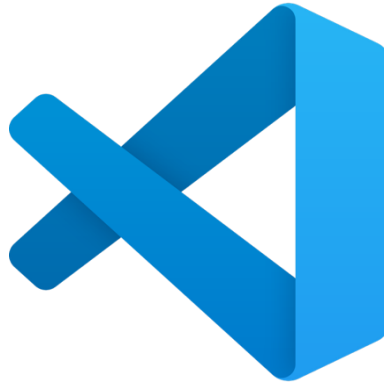
To classify $X = x$, we want to know k for which $P(Y = k|X = x)$ is largest.

Taking logs and discarding terms that do not depend on k we get the linear function of x :

$$\delta_k(x) = x \times \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$



What changes if X is not just a one-dimensional predictor?



Agenda.



- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- Naïve Bayes

What changes if X is not just a one-dimensional predictor?

Multivariate Gaussian distribution

We now extend the linear discriminant analysis to multiple predictors. We assume that $X = (X_1, \dots, X_p)$ is drawn from a multivariate gaussian distribution.

We write:

$$X \sim N(\mu, \Sigma)$$

With μ being the mean of X , a p -dimensional vector and Σ the $p \times p$ covariance matrix.

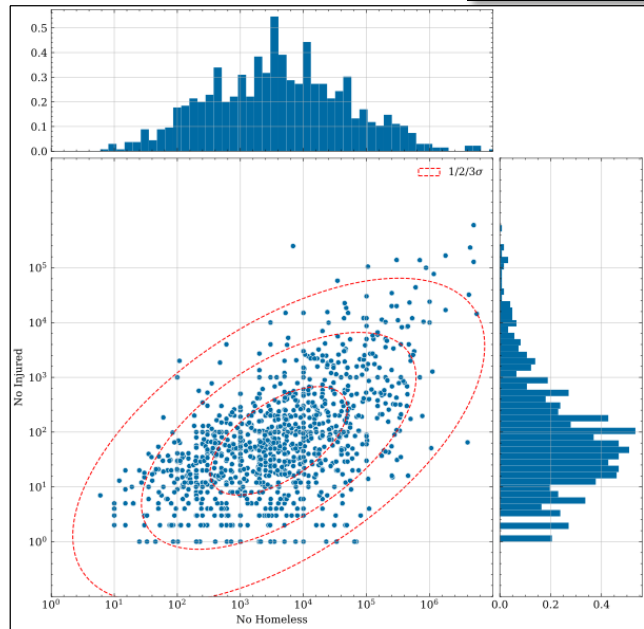


Figure 4.5: multivariate



So, what changes?

$$P(Y = k | X = x) = \frac{\pi_k \times f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

$$X \sim N\left(\mu = \begin{pmatrix} 3.61 \\ 1.84 \end{pmatrix}, \Sigma = \begin{pmatrix} 1.19 & 0.63 \\ 0.63 & 0.89 \end{pmatrix}\right)$$

What changes if X is not just a one-dimensional predictor?

Multivariate Gaussian distribution

$$P(Y = k | X = x) = \frac{\pi_k \times f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

The **gaussian density** function now has the form:

$$f_k(x) = \frac{1}{2\pi^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

And despite its complexity the **discriminant function** still has a linear form:

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$$

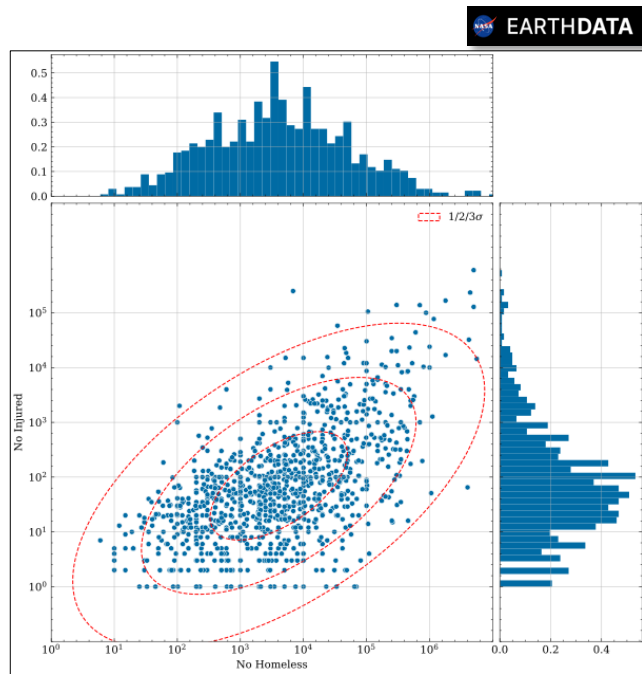


Figure 4.5: multivariate

$$X \sim N\left(\mu = \begin{pmatrix} 3.61 \\ 1.84 \end{pmatrix}, \Sigma = \begin{pmatrix} 1.19 & 0.63 \\ 0.63 & 0.89 \end{pmatrix}\right)$$

Different forms of Discriminant Analysis based on our assumptions.

Discriminant analysis

$$P(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

$\pi_k = P(Y = k)$ is the marginal or prior probability for class k . $\rightarrow$ its computation is straight forward.

$f_k(x) = P(X = x|Y)$ is the density for X in class $k \rightarrow$ that is the challenging part, where we discussed various assumptions.

Our assumptions:

Without any assumptions, **Bayes' theorem**.

If we assume $f_k(x)$ to be gaussian and Σ to be the same covariance matrix for each class, this leads to **linear discriminant analysis**.

With gaussians but different Σ_k in each class, we get **quadratic discriminant analysis**.

With $f_k(x) = \prod_{j=1}^p f_{jk}(x_j)$, conditional independence in each class, we get **naïve Bayes**. For Gaussian this mean Σ_k are diagonal.

What we need to estimate in $f_k(x)$:

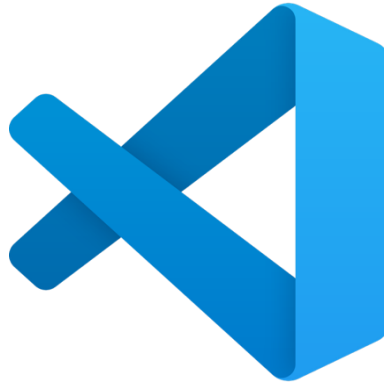
K p -dimensional density functions.

K p -dimensional mean vectors and one $p \times p$ -dimensional covariance matrix.

K p -dimensional mean vectors and K $p \times p$ -dimensional covariance matrix.



What changes?



- Recap on classification and regression
- Logistic regression
- Linear discriminant analysis
- Quadratic discriminant analysis
- **Naïve Bayes**

Why do we want to make the independent assumption for Naïve Bayes?

Discriminant analysis

$$P(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

With $f_k(x) = \prod_{j=1}^p f_{jk}(x_j)$, conditional independence in each class, we get **naïve Bayes**. For Gaussian this means Σ_k are diagonal.

What we need to estimate in $f_k(x)$:

K p -dimensional mean vectors
and K p -dimensional variance
vectors.

Motivation

In the case of a multivariate normal distribution, the association between the different predictors is summarized by the off-diagonal elements of the covariance matrix.

This association can be very hard to characterize and exceedingly challenging to estimate. If we want to compute the conditional probability $P(Y = k|X = x)$ for every observation $X = (X_1, \dots, X_p)$, we may not be able to observe every combination of observations and target classes. When p is large multivariate methods like **QDA** and even **LDA** break down.

By assuming that the p covariates are independent within each class, we completely eliminate the need to worry about the association between the p predictors.

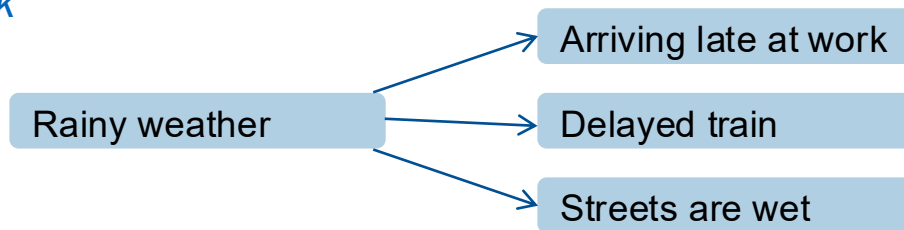
Probabilities of some events are not independent, e.g., the probability of a *delayed train (D)* and *arriving late at work (A)* are not independent (but positively correlated)

Probability that both D and A happen → $P(D \cap A) > P(D) P(A)$

However, these events may be **conditionally independent** if we have additional knowledge about the situation, e.g., that there is *rainy weather*

$$P(D \cap A | R) = P(D | R) P(A | R)$$

In this setting, the *rainy weather* explains the dependence between *delayed train* and *arriving late at work*



Task: Distinguish cats from dogs based on their size

- Classes cats and dogs: $Y \in \{c, d\}$
- Features height [cm], weight [kg]: $X_i = (h_i, w_i)$
- Training examples: $(X_i, Y_i) = (h_i, w_i, y_i)$ with 4 dogs and 12 cats



[1]

Given probabilities: $P(d) = \frac{4}{4+12} = 0.25$, $P(c) = \frac{12}{4+12} = 0.75$

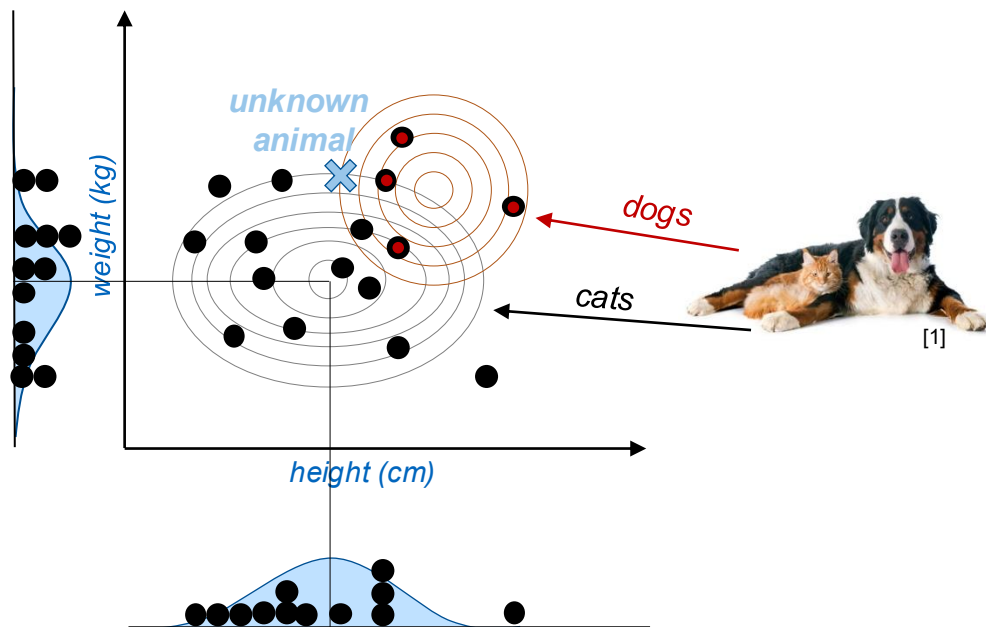
Model for dogs:

- Height ~ Gaussian with mean variance
- Weight ~ Gaussian $(\hat{\mu}_{w,d}, \hat{\sigma}_{w,d}^2)$
- Assume that height and weight are independent

$$\left\{ \begin{array}{l} \hat{\mu}_{h,d} = \frac{1}{4} \sum_{i:y_i=d} h_i \\ \hat{\sigma}_{h,d}^2 = \frac{1}{3} \sum_{i:y_i=d} (h_i - \hat{\mu}_{h,d})^2 \end{array} \right.$$

Model for cats:

- Same, using $(\hat{\mu}_{h,c}, \hat{\sigma}_{h,c}^2)$ and $(\hat{\mu}_{w,c}, \hat{\sigma}_{w,c}^2)$



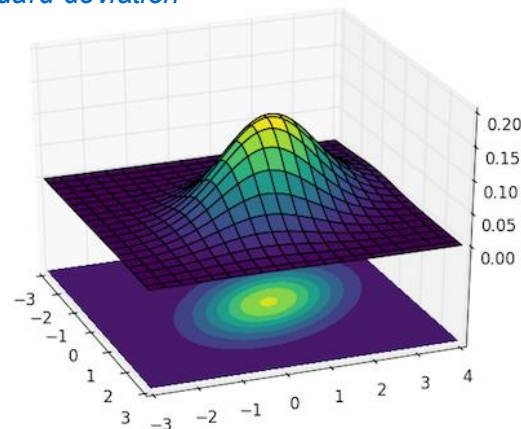
Recap: Gaussian Distribution

Probability density function:

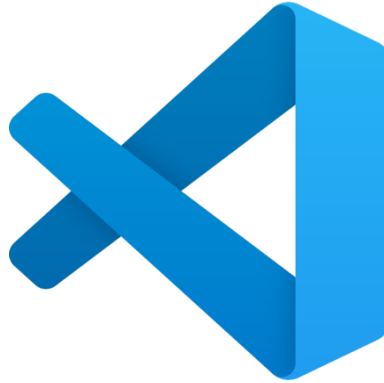
$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}$$

distribution mean (pointing to μ_k)

standard deviation (pointing to σ_k)



[2]



- Logistic regression is very popular for classification, especially when $K = 2$.
- LDA is useful when n is small, or the classes are well separated, and Gaussian assumption is reasonable. Also, when $K > 2$.
- Naïve Bayes is useful when p is very large.

Wrap up.



What is still **unclear** from this session?